Stellar Population Analysis Using Gaia DR3

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0.1. Key Terms and Definitions

Table 1: *Definitions of key astronomical terms used in this report.*

Term	Definition
Parsec (pc)	Unit of astronomical distance. $1\,\mathrm{pc} = 3.086 \times 10^{16}$ m ≈ 3.26 light-years.
Absolute Magnitude	The intrinsic brightness of a star as it would appear at a standard distance of 10 pc, removing the effect of distance.
Apparent Magnitude	The observed brightness of a star from Earth, affected by both intrinsic luminosity and distance.
Colour Index (BP−RP)	The difference in brightness between the blue (G_{BP}) and red (G_{RP}) Gaia bands. Higher values indicate cooler, redder stars.
Effective Temperature	The surface temperature of a star in Kelvin (K), which determines its colour and spectral classification.
Log Luminosity	Logarithm (base 10) of stellar luminosity relative to the Sun ($L_\odot$). Higher values indicate intrinsically brighter stars.
Spectral Class	Classification of stars by surface temperature: O, B, A, F, G, K, M (hottest to coolest).
M Dwarfs	Cool, faint, red stars of spectral class M ($T_{\mathrm{eff}} \approx 2400$–$3700$ K). The most numerous stars in the Galaxy but intrinsically faint.
Malmquist Bias	The tendency of magnitude-limited surveys to detect only the brightest stars at large distances, causing underrepresentation of intrinsically faint stars (**?**).
HR Diagram	Hertzsprung-Russell diagram: a plot of stellar luminosity vs effective temperature revealing stellar evolutionary stages (**?**).
Main Sequence	The principal band of stars on the HR diagram where core hydrogen fusion occurs. Stars spend ∼90% of their lifetimes on the main sequence.
Parallax	The apparent angular shift in a star's position due to Earth's orbital motion around the Sun, used by Gaia to measure heliocentric distances.

1. Results

1.1. Luminosity Function

Figure 1 presents the luminosity function , the distribution of absolute visual magnitudes for both stellar populations. An obvious offset can be noticed. The near-Earth population peaks at an absolute visual magnitude of approximately 10–11, corresponding to faint K and M dwarf stars. In contrast, the distant field population peaks at a significantly lower (brighter) magnitude of approximately 4–5, corresponding to solar-type G(stars moderate in size, temperature, brightness, similar to the sun) and subgiant stars(stars that have expanded massively after using up hydrogen in their core.).

This shift directly demonstrates Malmquist bias (**?**Butkevich et al., 2005): beyond 100 pc, Gaia's photometric sensitivity limit prevents detection of intrinsically faint stars such as M dwarfs, while only intrinsically bright stars remain detectable. The result is a sample that is biased toward luminous stars at large distances.

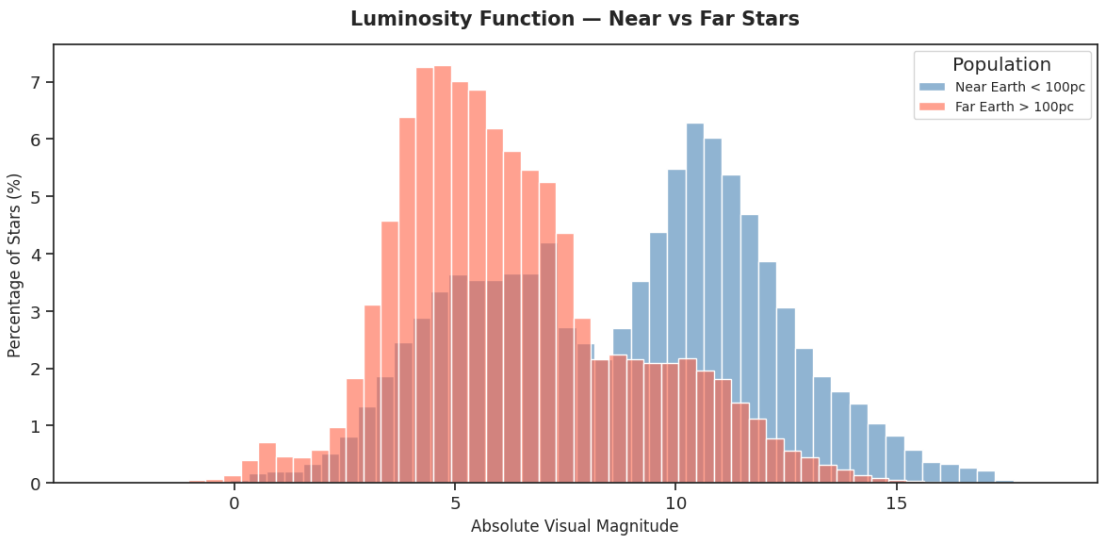

Figure 1: *Luminosity function showing the percentage distribution of absolute visual magnitude for near-Earth (blue, $d < 100$ pc) and distant field (red, $d > 100$ pc) stellar populations. The systematic shift toward brighter magnitudes in the distant population is direct evidence of Malmquist bias.*

1.2. Effective Temperature Distribution

Figure 2 shows the distribution of effective temperatures. The near-Earth population displays a great peak at approximately 3000–3500 K, consistent with the dominance of cool M dwarf stars in the solar neighbourhood. M dwarfs constitute approximately 70% of all stars in the Milky Way by number (Henry et al., 2018) and therefore are the most dominant stars in the nearby sample.

The distant field population peaks at a significantly higher temperature of approximately 4500–6000 K, corresponding to K and G type stars. The cool M dwarf peak is noticably suppressed. This temperature bias arises because M dwarfs ($M_V \sim 10$–16) are too faint to be reliably detected by Gaia beyond ∼100 pc, resulting in an apparent deficiency of cool stars in the distant sample.

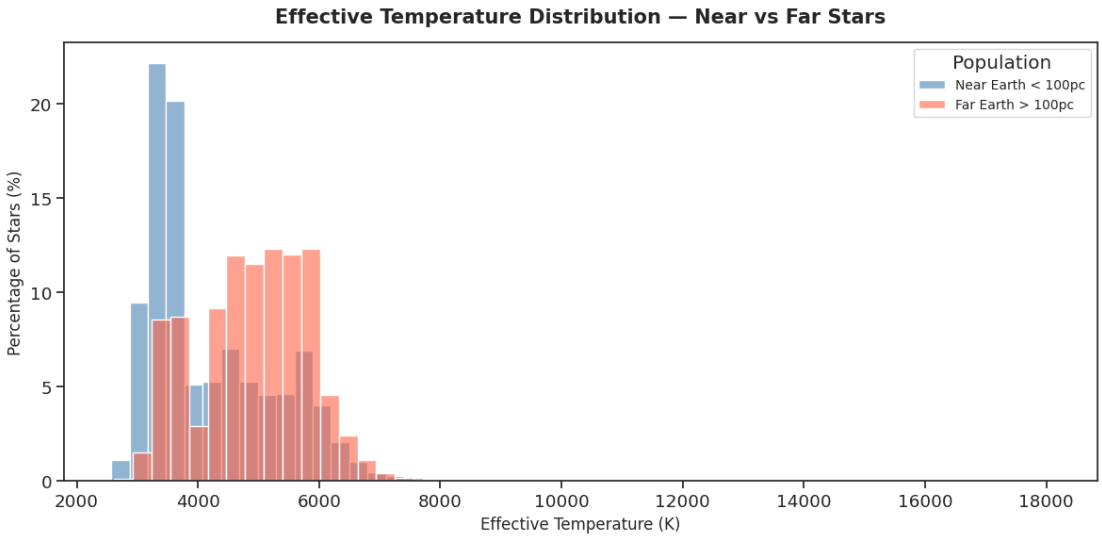

Figure 2: *Effective temperature distribution for near-Earth (blue) and distant field (red) stellar populations. The near-Earth population peaks at cool temperatures (∼3000 K) due to abundant M dwarfs, while the distant population peaks at warmer temperatures (∼5000 K), reflecting the loss of faint cool stars beyond the detection limit.*

1.3. Hertzsprung-Russell Diagram

Figure 3 presents Hertzsprung-Russell (HR) diagrams for both populations, plotting $\log(L/L_\odot)$ against $\log T_{\rm eff}$. In both panels, the main sequence is clearly visible as a diagonal band running from hot, luminous stars (upper left) to cool, faint stars (lower right). The main sequence represents the core hydrogen-burning phase and is the point where stars spend approximately 90% of their lifetimes (**?**).

A critical distinction between the two panels is the extent of the main sequence at its faint, cool end. The near-Earth HR diagram extends to very low luminosities ($\log L \approx -4$), populated by M dwarfs. In contrast, the distant field HR diagram is truncated at the faint end; the lower main sequence is sparsely populated because cool red dwarf stars are undetectable at large distances. This truncation is a direct observational signature of Malmquist bias.

It is important to note that the main sequence visible in the distant field population does *not* represent a same-age stellar population. Rather, stars of vastly different ages and distances converge on the main sequence simply because it is the longest lived evolutionary

phase for all stellar masses fundamentally different from a cluster main sequence where all stars share the same age, distance, and chemical composition (**?**).

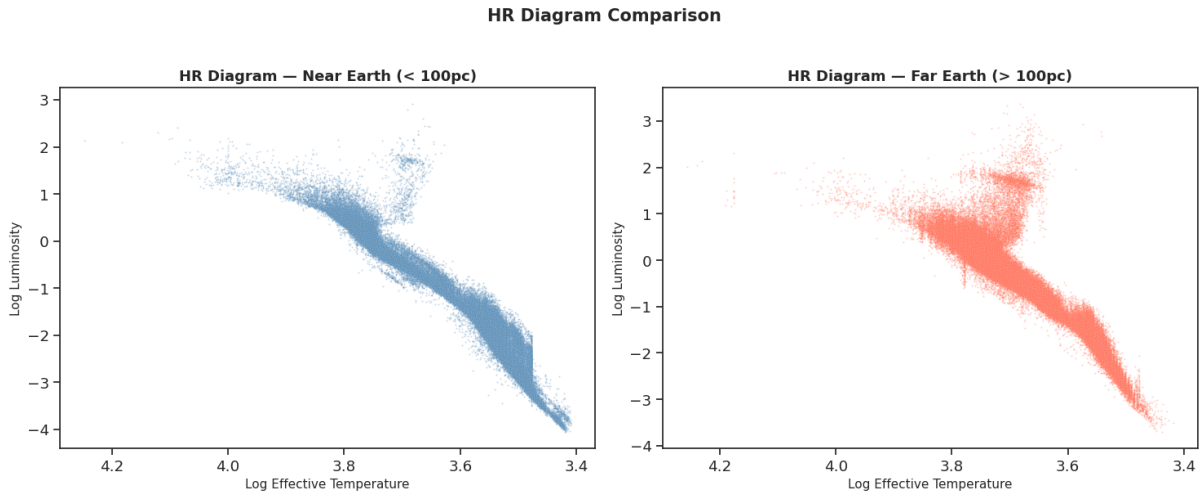

Figure 3: *Hertzsprung-Russell diagrams for near-Earth (left, blue, $d < 100$ pc) and distant field (right, red, $d > 100$ pc) stellar populations. Both show a clear main sequence, but the distant population is truncated at the faint, cool end due to the Gaia detection limit..*

1.4. Spectral Class Distribution

Figure 4 quantifies the spectral class composition of both populations as percentage fractions. In the near-Earth population, M-type stars constitute approximately 50% of all detected stars consistent with their known numerical dominance in the local Universe (Bochanski et al., 2010). K-type stars account for approximately 29%, while G and F type stars are less common.

In contrast, the distant field population shows a dramatic reduction in M-type stars to approximately 16%, while K-type stars rise to ∼43% and G-type stars to ∼32%. The systematic underrepresentation of M dwarfs at large distances and the corresponding overrepresentation of intrinsically brighter K and G type stars reflects the magnitude limit of the survey rather than any genuine physical difference between the two populations.

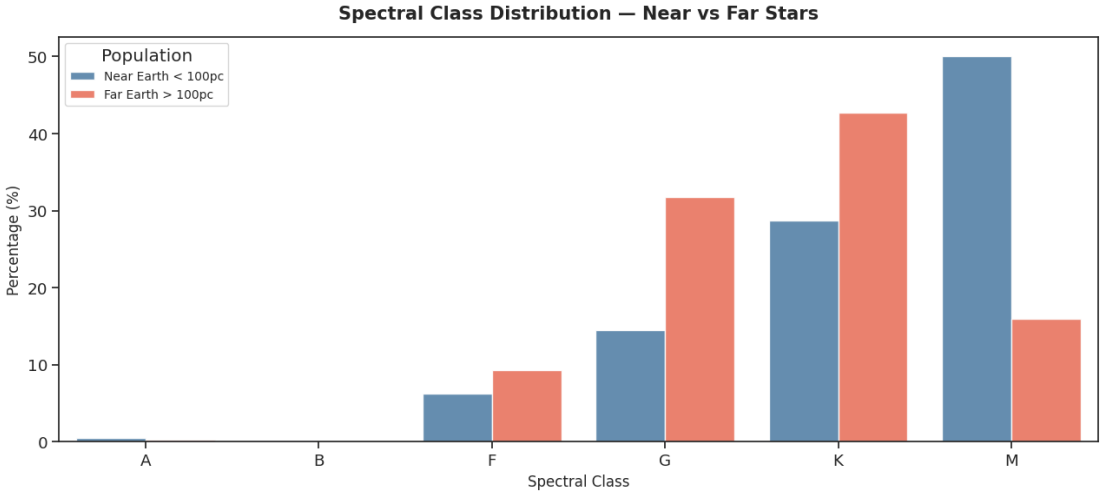

Figure 4: *Spectral class percentage distribution for near-Earth (blue) and distant field (red) populations. M dwarfs dominate the nearby sample (∼50%) but are strongly underrepresented in the distant sample (∼16%), providing direct quantitative evidence of Malmquist bias.*

References

Bochanski, J. J., Hawley, S. L., Covey, K. R., et al. 2010, *The Astronomical Journal*, 139, 2679. doi:10.1088/0004-6256/139/6/2679

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